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Chapter 1
Assuming the Role of the Systems Analyst

Systems Analysis and Design

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Major Topics

- Information systems
- Phases of analysis and design
- System maintenance
- CASE tools

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Information

- Information is an organizational resource which must be managed as carefully as other resources
- Costs are associated with information processing
- Information processing must be managed to take full advantage of its potential

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Categories

- Information systems fall into one of the following eight categories:
  - Transaction processing systems (TPS)
  - Office automation systems (OAS)
  - Knowledge work systems (KWS)
  - Management information systems (MIS)
  - Decision support systems (DSS)
  - Expert systems (ES)
  - Executive support systems (EES)

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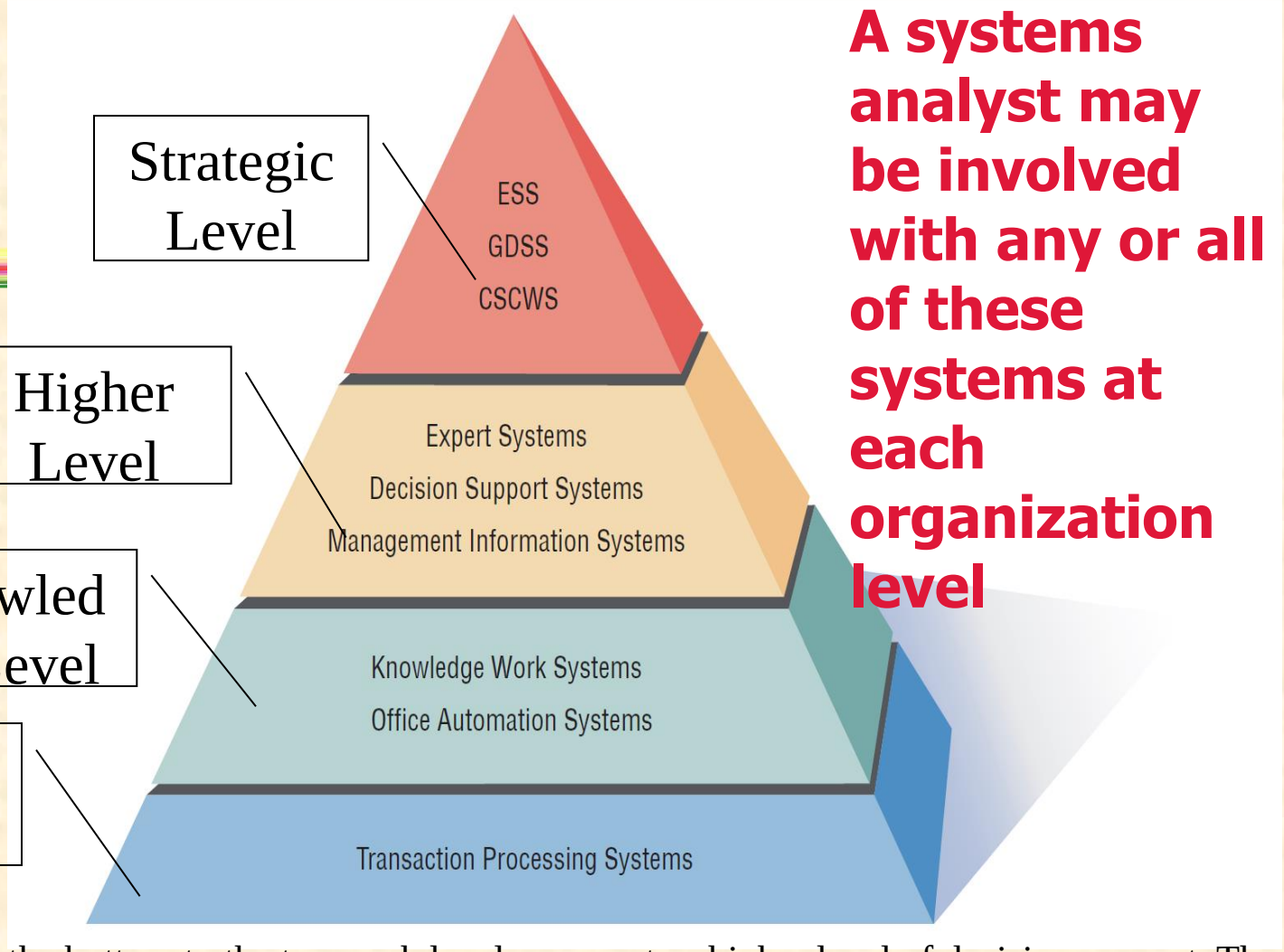

A systems analyst may be involved with any or all of these systems at each organization level

As we move from the bottom to the top, each level represents a higher level of decision support. The bottom level or organizational level supported by TPS provides very structured decision support while the top level or strategic level supported by ESS, GDSS and CSCWS provides semi-structured and unstructured decision support.

Transaction processing systems (TPS)

- TPS are computerized information system developed to process large amounts of data for routine business transactions such as payroll and inventory.
- TPS function at the operational level of the organization.

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Office automation systems (OAS) and Knowledge work systems (KWS)

- At the knowledge level of the organization.
- OAS support data workers who don't create new knowledge but use information for analysis to transform data or manipulate it then share it .
- OAS include word processing , spreadsheets, electronic scheduling, and communication through voice mail, E-mail

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Office automation systems (OAS) and Knowledge work systems (KWS)

- KWS support professional workers such as scientists, engineers, and doctors by aiding them to create new knowledge and contribute it to their organization.

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MIS Management Information System

- Don’t replace TPS. Rather all MIS include TPS
- MIS are computerized information systems that support a broader spectrum of organizational tasks.
- MIS output information is used in decision making.

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Decision support systems (DSS)

- A higher level class of computerized information system.
- DSS is similar to MIS in that they both depend on a database as a source of data.
- DSS departs from MIS in that it emphasizes the support of decision making in all of its phases.

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Expert systems (ES)

- ES are very special class of IS.
- It also called knowledge based system.
- It captures and uses the knowledge of an expert for solving a particular problem.

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ESS Executive

- ESS helps executives to organize their interactions with the external environment.
- ESS rely on the information generated by TPS and MIS.
- Support strategic level.

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New Technologies

- New technologies are being integrated into traditional systems
  - Ecommerce uses the Web to perform business activities
  - Enterprise Resource Planning (ERP) has the goal of integrating many different information systems within the corporation
  - Wireless and handheld devices, including mobile commerce (Mcommerce)

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Ecommerce and Web Systems

- **Benefits**
  - Increasing user awareness of the availability of a service, product, industry, person, or group
  - The possibility of 24-hour access for users
  - Improving the usefulness and usability of interface design
  - Creating a system that can extend globally rather than remain local, thus reaching people in remote locations without worry of the time zone in which they are located

Enterprise Resource Planning Systems (ERP)

- Performs integration of many information systems existing on different management levels and within different functions
- Instituting ERP requires enormous commitment and organizational change.
- Example: Oracle

Wireless Systems

- System analyst may be asked to design standard or wireless communication networks that integrate voice, video and email into organizational intranets or industry extranets
- System analyst may also be asked to develop intelligent agents

Intelligent agents are software that can assist users with tasks in which the software learns preferences from of users over time and then acts on those preferences.

- Wireless communication is referred as m-commerce (mobile commerce)

Nature of Analysis and Design

- Systems analysis and design is a systematic approach to identifying problems, opportunities, and objectives; analyzing the information flows in organizations; and designing computerized information systems to solve a problem

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Systems Analyst

- Systems analysts act as
  - Outside consultants to businesses
  - Supporting experts within a business
  - As change agents. You are an agent of change whenever you perform any of the activities in the SDLC
- Analysts are problem solvers, and require communication skills
- Analysts must be ethical with users and customers

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Systems Development Life Cycle

- The systems development life cycle is a systematic approach to solving business problems
- It is divided into seven phases
- Each phase has unique activities

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Phase 1

- Identifying
  - Problems
  - Opportunities: are situations that the analyst believes can be improved through the use of computerized information system
  - Objectives: the analyst must discover what the business is trying to do.

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Phase 1

- Personnel involved
  - Analyst
  - User management
  - Systems management

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Phase 2

- Determining information requirements
  - Interview management, operations personnel
  - Gather systems/operating documents
  - Use questionnaires
  - Observe the system and personnel involved
- Learn the who, what, where, when, and how, and the why for each of these

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Phase 2

- Personnel involved
  - Analyst
  - User management
  - User operations workers
  - Systems management

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Phase 3

- Analyzing system needs
  - Create data flow diagrams
  - Document procedural logic for data flow diagram processes
  - Complete the data dictionary
  - Make semistructured decisions
  - Prepare and present the system proposal
  - Recommend the optimal solution to management

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Phase 3

- Personnel involved
  - Analyst
  - User management
  - Systems management

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Phase 4

- Designing the recommended system
  - Design the user interface
    - Design output
    - Design input
  - Design system controls
  - Design files and/or database
  - Produce program specifications
  - Produce decision trees or tables

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Phase 4

- Personnel involved
  - Analyst
  - System designer
  - User management
  - User operations workers
  - Systems management

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Phase 5

- Developing and documenting software
  - Design computer programs using structure charts, Nassi-Schneiderman charts, and pseudocode
  - Walkthrough program design
  - Write computer programs
  - Document software with help files, procedure manuals, and Web sites with Frequently Asked Questions

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Phase 5

- Personnel involved
  - Analyst
  - System designer
  - Programmers
  - Systems management

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Phase 6

- Testing and maintaining the system
  - Test and debug computer programs
  - Test the computer system
  - Enhance system

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Phase 6

- Personnel involved
  - Analyst
  - System designer
  - Programmers
  - Systems management

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Phase 7

- Implementing and evaluating the system
  - Plan conversion
  - Train users
  - Purchase and install new equipment
  - Install system
  - Review and evaluate system

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Phase 7

- Personnel involved
  - Analyst
  - System designer
  - Programmers
  - User management
  - User operations workers
  - Systems management

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System Maintenance

- System maintenance is enhancing existing software
- Over time the cost of continued maintenance will be greater than that of creating an entirely new system.

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System Enhancements

- Systems are enhanced for the following reasons:
  - Adding additional features to the system
  - Business and governmental requirements change over time
  - Technology, hardware, and software are rapidly changing

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CASE Tools

- Computer Aided Software Engineering
- CASE tools are automated, computer-based software packages for systems analysis and design
- Four reasons for using CASE tools are:
  - To increase analyst productivity
  - Facilitate communication among analysts and users
  - Providing continuity between life cycle phases
  - To assess the impact of maintenance

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CASE Tool Categories

- CASE tools may be divided into several categories
  - Upper CASE (also called front-end CASE) tools, used to perform analysis and design
  - Lower CASE (also called back-end CASE). These tools generate computer language source code from CASE design
  - Integrated CASE, performing both upper and lower CASE functions

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Upper CASE

- Upper CASE tools
  - Create and modify the system design
  - Store data in a project repository
  - The repository is a collection of records, elements, diagrams, screens, reports, and other project information
  - These CASE tools model organizational requirements and define system boundaries

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Lower CASE

- Lower CASE tools generate computer source code from the CASE design
- Source code may usually be generated in several languages

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Advantages of Generating Code

- Time to develop new systems decreases
- The time to maintain generated code is less than to maintain traditional systems
- Computer programs may be generated in more than one language
- Generated code is free from program coding errors

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Reverse Engineering

- Reverse engineering is generating the CASE design from computer program code
- Source code is examined, analyzed, and converted into repository entities

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Reverse Engineering Produces

- Reverse engineering produces (depending on the tool set used)
  - Data structures and elements, describing the files, records, and field
  - Screen designs
  - A structure chart showing the hierarchy of the modules in the program
  - Database design and relationships

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Object-Oriented Analysis and Design

- A new and very different approach to systems analysis and design is O-O
- Object-oriented (O-O) analysis and design is used to build object-oriented programs
- This includes not only data but the instructions about operations that manipulate the data

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Types of Object-Oriented Analysis and Design

- There are three types of object-oriented analysis and design:
  - Object-oriented analysis (OOA)
  - Object-oriented design (OOD)
  - The Unified Modeling Language (UML), a standardized object-oriented modeling language

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Nature of Analysis and Design

- Structured analysis and design provides a systematic approach to developing systems and is cyclic in nature
- Analysis and design errors detected in the later phases of the systems development life cycle cost more to fix than if detected in earlier phases

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